

IOWA COUNTY AP, WI, USA

WMO: 726507

Lat: 42.887N

Lon: 90.236W

Elev: 357

StdP: 97.11

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-17.9	-26.0	0.4	-20.4	-22.7	0.5	-17.2	11.2	-4.2	10.1	-5.8	3.3	290	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	32.0	23.1	30.1	22.6	28.1	21.6	25.6	29.0	24.3	28.1	23.3	27.0	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.9	27.7	22.9	18.4	26.5	22.3	17.8	25.9	81.2	29.6	75.3	28.6	71.4	26.9	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.9	8.0	-25.1	33.9	3.4	1.6	-27.6	35.1	-29.6	36.0	-31.5	36.9	-34.0	38.0		
			-25.1	26.9	3.0	1.2	-27.2	27.8	-29.0	28.5	-30.7	29.2	-32.9	30.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-6.2	-4.3	2.1	8.9	15.2	20.3	22.5	21.3	17.4	10.4	3.5	-3.3
	DBStd	11.29	6.60	6.41	6.43	5.27	4.80	3.62	3.15	2.94	4.43	5.13	5.46	6.36
	HDD10.0	1933	502	402	258	81	10	0	0	0	2	59	204	414
	HDD18.3	3775	760	635	505	286	121	22	4	8	68	252	443	672
	CDD10.0	1582	0	1	12	48	171	308	386	349	225	71	10	1
	CDD18.3	383	0	0	1	2	24	79	132	99	41	4	0	0
	CDH23.3	3030	0	0	6	29	206	632	1102	701	321	34	0	0
	CDH26.7	832	0	0	0	4	48	169	339	181	86	4	0	0
Wind	WSAvg	3.6	3.9	4.0	4.2	4.6	3.9	3.1	2.7	2.5	3.0	3.6	4.0	3.9
Precipitation	PrecAvg	969	32	36	55	95	108	147	114	118	88	66	62	41
	PrecMax	1227	64	73	143	183	264	265	239	437	183	141	189	115
	PrecMin	735	6	2	11	35	21	27	46	39	16	16	7	5
	PrecStd	151	16	20	36	43	52	66	47	84	42	39	47	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	13.9	23.0	27.1	31.2	32.5	34.9	32.6	32.1	27.2	20.2	14.1
		MCWB	5.8	10.6	16.6	16.2	20.5	23.3	23.1	22.1	21.9	19.3	14.1	13.1
	2%	DB	6.2	8.9	17.9	22.8	27.6	30.2	32.0	30.2	29.1	23.8	17.0	10.1
		MCWB	3.9	6.9	13.2	14.6	18.7	22.0	24.5	22.8	21.6	17.5	12.6	8.1
	5%	DB	3.0	6.3	14.8	20.2	26.0	28.2	30.0	28.1	27.2	21.3	14.0	7.0
		MCWB	2.1	4.1	10.7	13.1	17.7	21.5	23.3	21.9	20.1	16.3	10.6	5.4
	10%	DB	2.3	3.6	11.2	17.6	22.9	27.2	28.0	27.3	24.9	18.7	12.1	4.0
		MCWB	1.5	2.3	7.7	11.4	16.7	20.7	22.5	21.2	18.8	15.2	9.1	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	12.0	17.2	18.4	22.6	26.5	27.8	25.7	24.6	23.1	15.7	13.7
		MCDB	7.8	12.9	22.6	23.9	28.2	28.8	30.2	29.1	28.8	25.8	18.5	14.8
	2%	WB	3.9	7.3	14.2	16.3	20.8	24.6	26.3	24.4	23.0	19.6	13.4	8.0
		MCDB	5.8	8.5	17.1	20.6	25.3	28.5	29.5	28.2	26.6	21.5	16.0	9.0
	5%	WB	2.5	4.2	11.0	14.5	19.4	23.1	24.9	23.5	21.6	17.2	11.6	5.2
		MCDB	3.5	5.9	13.8	19.2	24.0	27.0	28.4	27.1	25.6	19.7	13.8	6.9
	10%	WB	1.3	2.5	7.7	12.1	17.9	21.8	23.6	22.4	20.2	14.9	8.7	3.0
		MCDB	2.1	3.6	10.4	16.4	21.9	25.5	27.2	25.8	23.8	18.4	11.2	4.0
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.9	8.6	10.2	10.3	9.8	9.8	9.7	10.7	9.6	8.0	6.9
		MCDBR	7.9	9.4	12.4	14.1	12.4	11.4	10.8	11.0	12.6	12.7	11.4	8.4
		MCWBR	6.5	7.2	7.8	8.1	6.5	6.0	5.3	5.4	6.1	7.4	8.0	7.1
	5% WB	MCDBR	7.8	9.2	11.3	12.4	10.7	10.1	9.5	9.6	10.7	10.9	10.8	8.3
		MCWBR	6.5	7.1	8.2	8.4	6.7	6.5	6.2	5.7	6.3	8.5	8.6	7.5
Clear-Sky Solar Irradiance	taub	0.274	0.300	0.321	0.365	0.392	0.412	0.419	0.404	0.372	0.331	0.292	0.274	
	taud	2.392	2.322	2.409	2.315	2.291	2.270	2.251	2.295	2.366	2.480	2.517	2.439	
	Ebn at Noon	871	898	924	901	882	862	850	852	856	856	849	840	
	Edh at Noon	77	98	103	123	129	133	134	124	107	84	69	68	
All-Sky Solar Radiation	RadAvg	1.76	2.79	3.79	4.58	5.41	6.04	6.08	5.30	4.36	2.82	1.85	1.36	
	RadStd	0.21	0.29	0.42	0.38	0.39	0.46	0.35	0.35	0.35	0.35	0.15	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page